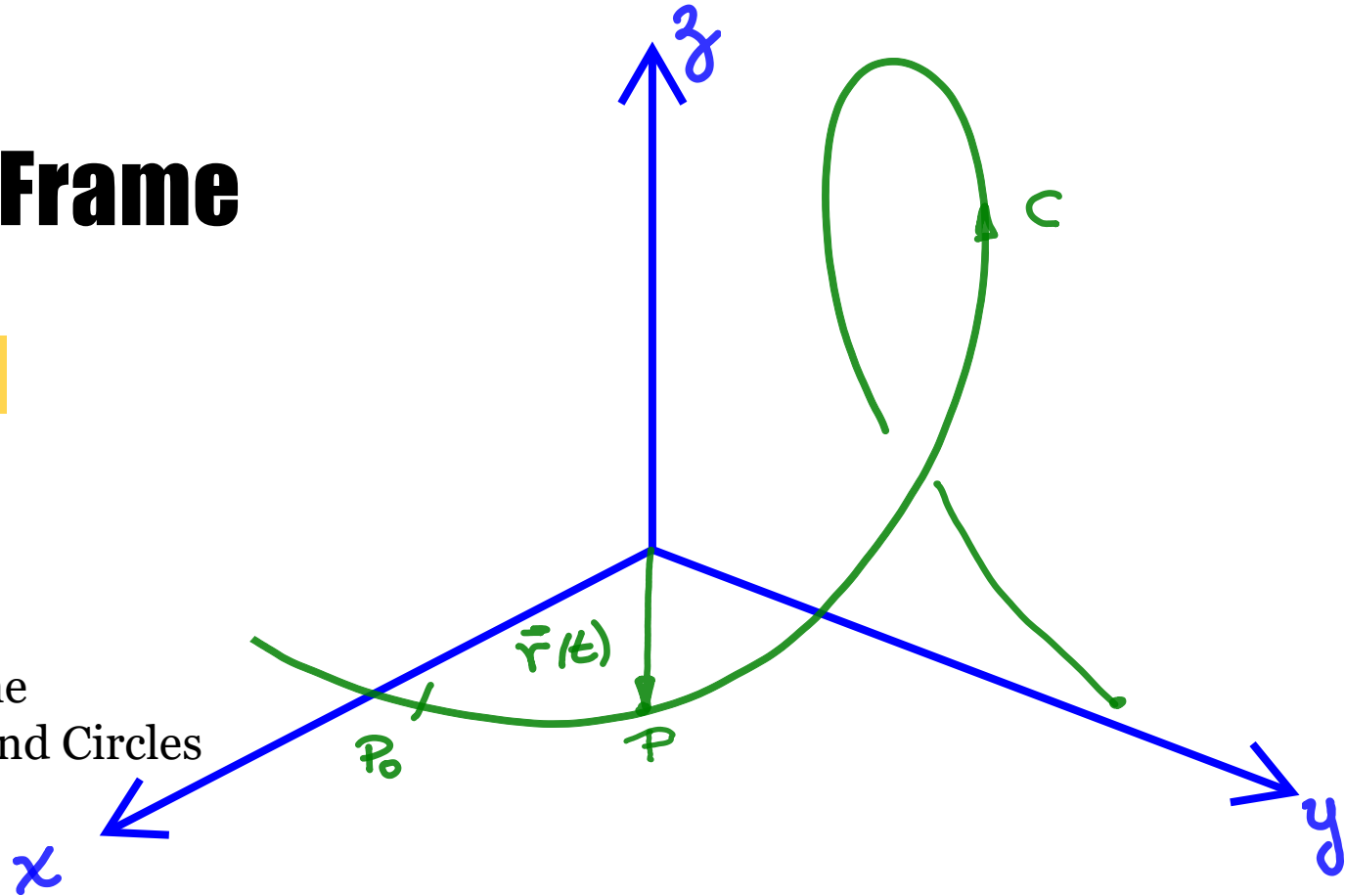


The Frenet Frame

Trim Ch 11, S.11.12

Outline:

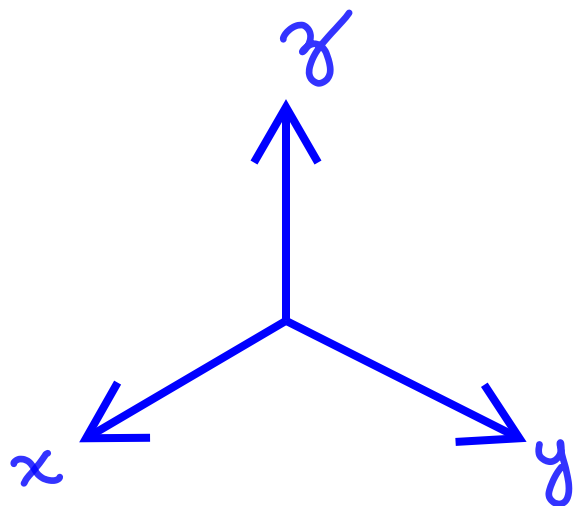
1. The Frenet Frame
2. Related Planes and Circles



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Principal Unit Normal Vector

Zooming in a piece of the curve C



The principal unit normal vector $\hat{N}$ is one of those vectors orthogonal to $\hat{T}$ but points in the direction in which the curve is turning (concave side of the curve).



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Principal Unit Normal Vector

Let us relate the principal unit normal vector $\hat{N}$ to $\hat{T}(s)$



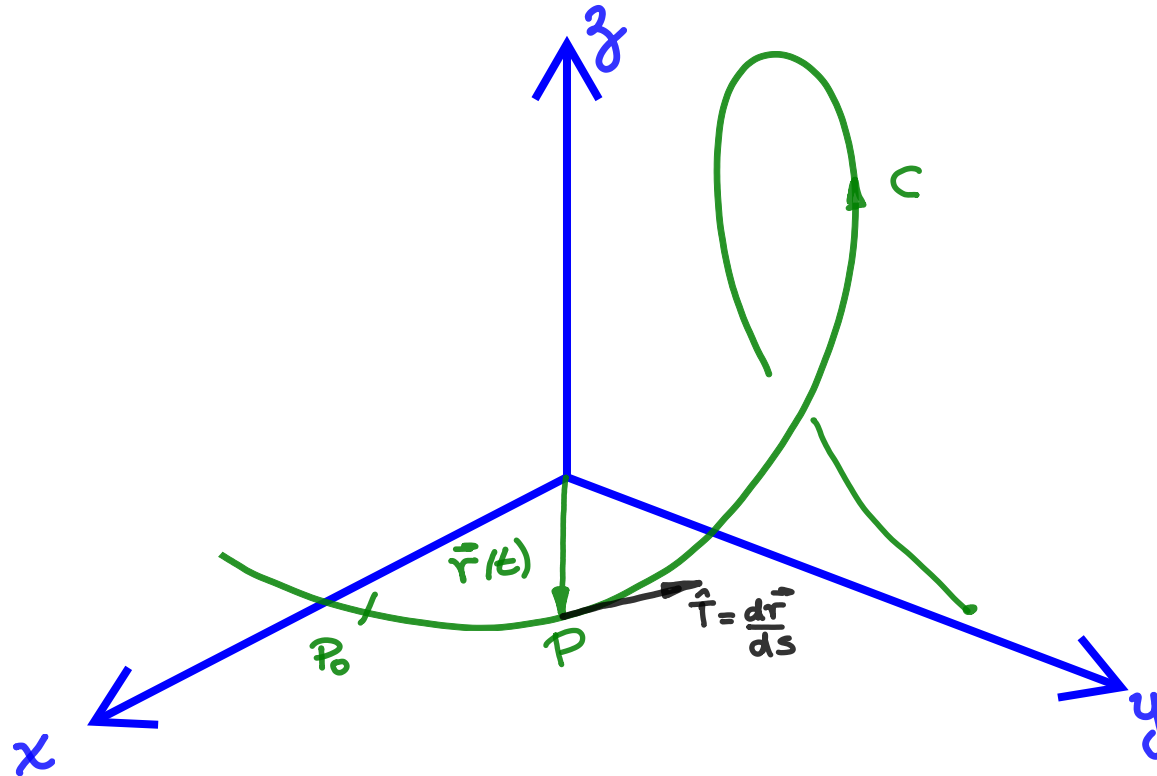
Principal Unit Normal Vector

The principal unit normal vector $\hat{N}$ for a curve given in terms of the parameter “ t ” can be calculated as,

Note that this expression of $\hat{N}$ does not need parametrization as a function of s , and it does not require the value of the curvature!



Unit Binormal Vector



The tendency of the motion to “twist” out of plane created by $\hat{T}$ and $\hat{N}$ in the direction perpendicular to it is define by,

$$\hat{B} = \hat{T} \times \hat{N}$$

and $\hat{B}$ is the Binormal Vector.



The Frenet Frame

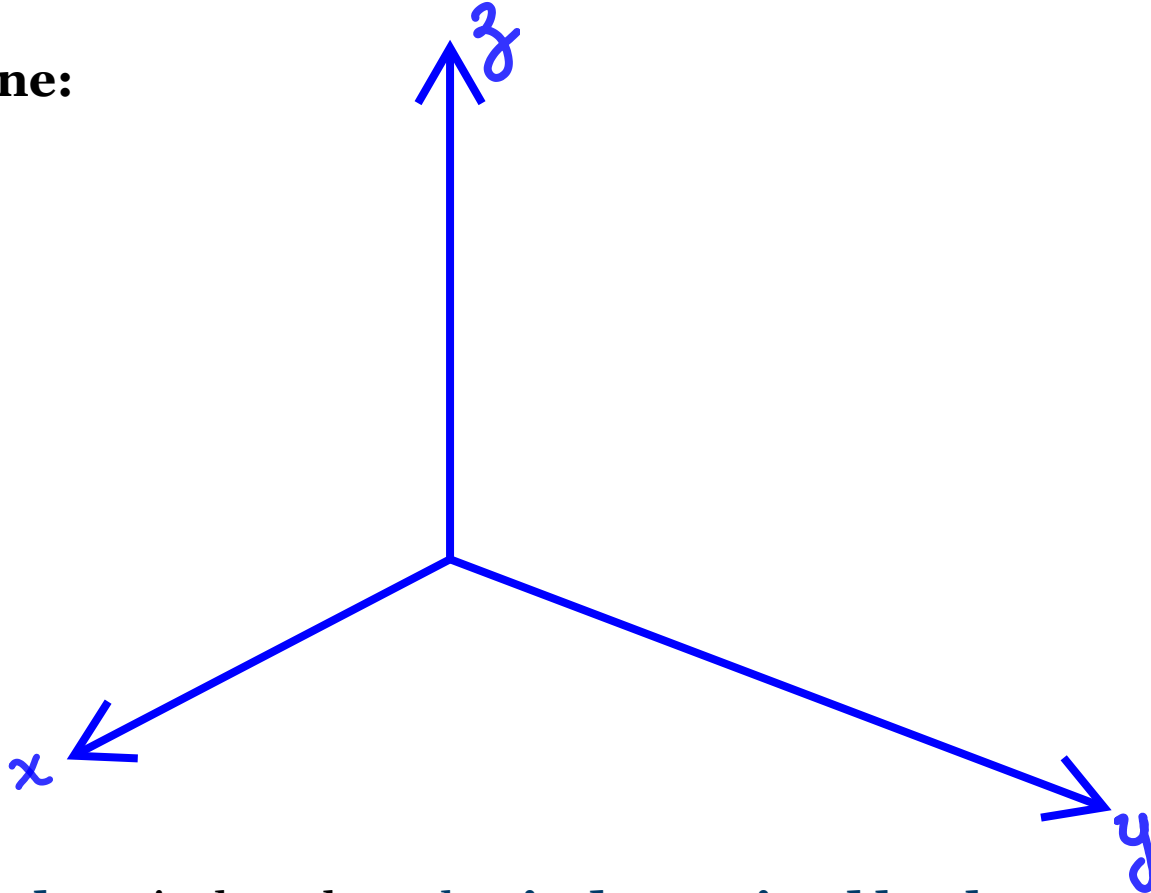
- Together $\hat{T}$ and $\hat{N}$ and $\hat{B}$ define a **moving right-handed vector frame** called **The Frenet** (Fre-Nay) **Frame**.
- This vector frame plays a significant role in calculating paths of particles moving through space
 - The Frenet Frame helps robots and self-driving cars **think like drivers**, focusing on the road ahead and how to move along it, rather than worrying about the whole map (like GPS coordinates).

Example:

Express a vector $\vec{F}$ in terms of the Frenet Frame?

Important Related Planes and Circles

Normal Plane:



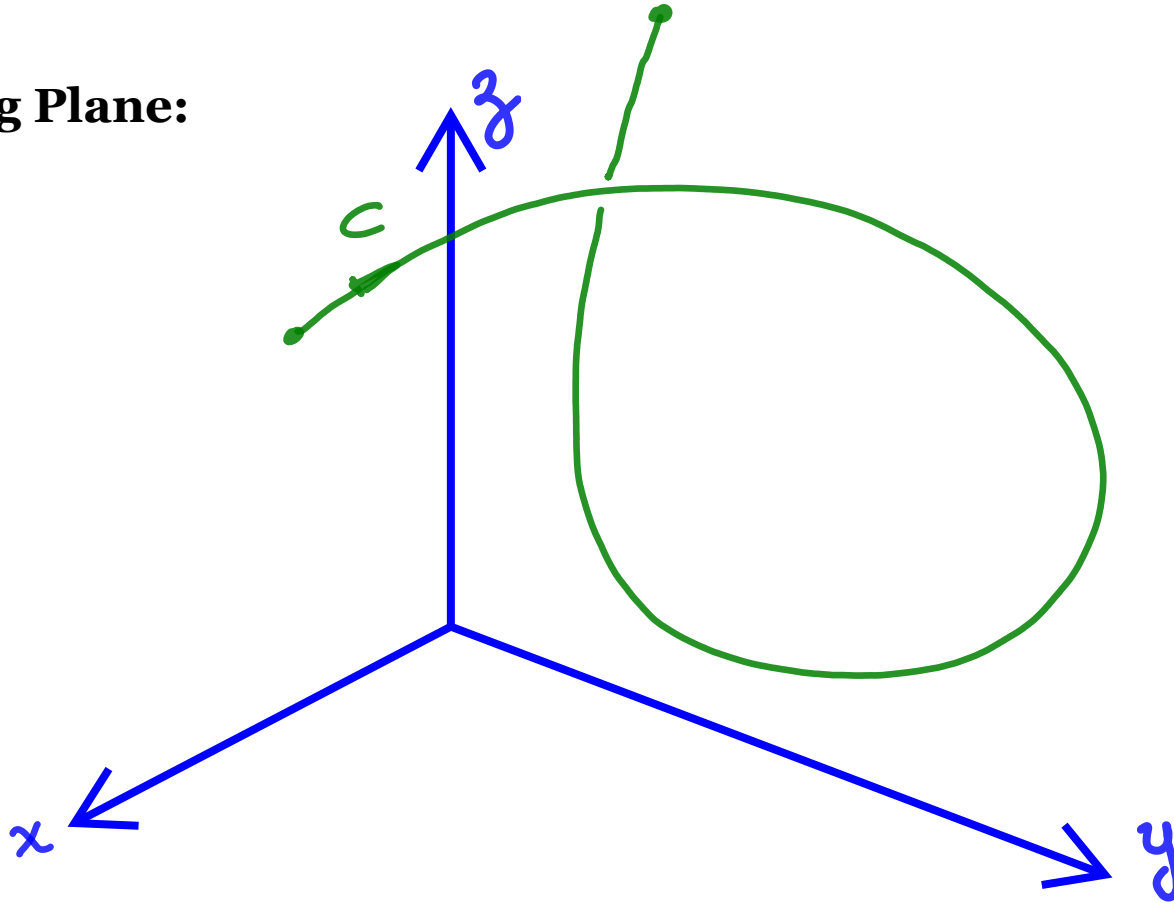
The **normal plane** is that plane that is determined by the vectors $\hat{N}$ and $\hat{B}$ at a point P on a curve C .



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Important Related Planes and Circles

Osculating Plane:

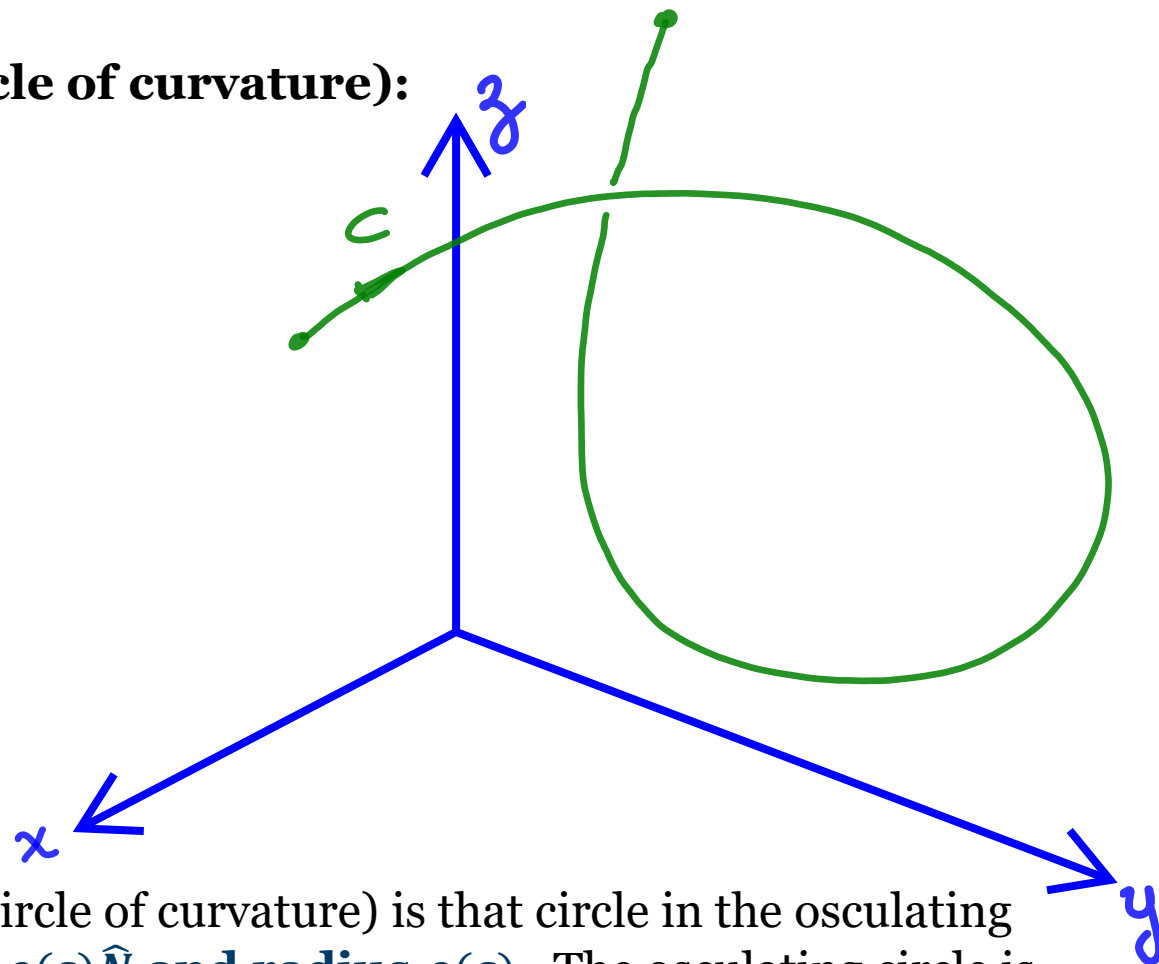


The **osculating plane** is that plane that is determined by the vectors $\hat{T}$ and $\hat{N}$ at a point P on a curve C .



Important Related Planes and Circles

Osculating Circle (or circle of curvature):



The **osculating circle** (or circle of curvature) is that circle in the osculating plane with **center at $\vec{r}(s) + \rho(s)\hat{N}$** and **radius $\rho(s)$** . The osculating circle is the best circle that approximates the curve at P .



Summary

- The principal unit vector $\hat{N}$ points toward the concave side of the curve
- The tendency of the motion to “twist” out of plane created by $\hat{T}$ and $\hat{N}$ in the direction perpendicular to it is define by $\hat{B}$
- Together $\hat{T}$ and $\hat{N}$ and $\hat{B}$ define a moving right-handed vector frame called **The Frenet** (Fre-Nay) **Frame**, which simplifies calculations during path planning.
- Important related planes and circles for modeling the path of a particle are the normal plane, the osculating plane and the osculating circle.





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Thank you